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Upper Ordovician palaeogeography of the Bohemian Massif: implications for Armorica

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SUMMARY

The palaeogeographic position of Armorica in the upper Ordovician is still rather unclear, due mainly to the lack of reliable palaeomagnetic data. To help resolve this, a palaeomagnetic study of Caradocian and Ashgillian sediments and volcanics of the Barrandian Basin in the central Bohemian Massif (Czech Republic) has been carried out. A total of 29 sites were sampled, including four from a diabase sill intruded into Hirnantian-age sediments, thus providing a baked contact test. Three components of magnetization, labelled A, B and C, have been resolved, in addition to a low-temperature component which corresponds to the present-day direction of the local geomagnetic field ($D/I = 360°/67°$). Component A (southerly and shallow before bedding correction) is only sporadically present, and corresponds to a remagnetization event of Permo–Carboniferous age which is well known throughout Hercynian Europe. Component B, present as stable endpoints in the sill, and as a secondary component in the host-rock sandstones, yields an overall mean direction of 360°/39°, $\alpha_{95} = 9.3°$, $k = 52.3$ *in situ*, and 355°/21°, $\alpha_{95} = 9.3°$, $k = 52.3$ after bedding correction. Component C, identified as stable endpoints in sedimentary and volcanic rocks, passes both a fold test and a baked contact test and is interpreted as being the primary direction of magnetization. The overall mean direction for C is 183°/−59°, $\alpha_{95} = 9.5°$, $k = 30.3$ (nine sites) after bedding correction which corresponds to a palaeopole position of 80°S; 360°E. This translates into palaeolatitudes of 40°S for the area studied in latest Ordovician times, but indicates large amounts of rotation (up to 170°) of the Bohemian Massif before final consolidation of Hercynian Europe. Although such large amounts of rotation are difficult to accommodate, they are consistent with palaeomagnetic results obtained from lower Ordovician and upper Silurian rocks elsewhere in the Barrandian Basin.

If the Bohemian Massif is interpreted as being an integral part of the Armorican microplate, then these results indicate that prior to Caradoc times, Armorica rifted away from the northern margin of Gondwana, and that by Ashgillian times it was located at intermediate palaeolatitudes.

Key words: Armorica, Bohemian Massif, palaeomagnetism, upper Ordovician.

INTRODUCTION

The early Palaeozoic drift history of Gondwana and the microplates derived from its northern margin, Avalonia and Armorica, are major factors controlling the development of Hercynian Europe. Recent palaeomagnetic studies in Baltica and Avalonia have resulted in fairly well constrained palaeodrift histories for these elements during lower Palaeozoic times. Avalonia rifted off the northern margin of Gondwana in Arenig/Llanvirn times and drifted northwards, whilst Baltica moved from fairly high palaeolatitudes to a more equatorial position during mid to upper Ordovician times (Torsvik *et al.* 1992). These two elements (Baltica and Avalonia) then collided in early Silurian times with closure of the intervening Tornquist Sea (Channell *et al.* 1992; Torsvik *et al.* 1993).

The scenarios for Armorica, constituting the Iberian, Armorican and Bohemian Massifs, and Gondwana are less clear (Bachtadse *et al.* 1995). Whilst it is now clear from palaeomagnetic evidence that from early Ordovician to late Silurian times Armorica drifted from a high peripolar position (Duff 1979; Perroud *et al.* 1986; Cogné 1988; Tait, Bachtadse & Soffel 1994a) to more temperate palaeolatitudes (Tait, Bachtadse & Soffel 1994b), the geodynamic

history of this terrane assemblage from the Llanvirn to Wenlock, and its relationship to Avalonia and Baltica, remains enigmatic. Faunal evidence is rather unclear due mainly to the widespread late Ordovician transgression which caused a breakdown in faunal provinciality and the deposition of black graptolite shales throughout Armorica in the Silurian. However, from mid-Caradoc times, the warmer-water Baltic faunas were able to colonize the shelf regions of both Avalonia and Armorica, whilst the faunas of northern Gondwana remained of low diversity and endemic to Gondwana (Cocks & Fortey 1988).

A late Ordovician period of continental glaciation, with the ice sheet centred over northern Africa, is now well established (Beuf *et al.* 1971), and recent isotopic and bathymetric evidence indicates it was very short-lived, being restricted essentially to the Hirnantian (latest Ashgill) (Brenchley *et al.* 1994). The presence of Hirnantian glaciomarine deposits in France, Bohemia and Iberia (Storch 1990; Brenchley *et al.* 1991) would tend to suggest fairly high palaeolatitudes and proximity of Armorica to Gondwana. However, it has been demonstrated that the diamictites of the Bohemian Massif were deposited from seasonal or floating ice (Brenchley *et al.* 1991), and thus do not necessitate proximity of this region to Gondwana. In the Barrandian Basin, the diamictites are overlain by the Kosov Formation containing the widely recognized *Hirnantia* fauna. Although the ancestors to members of this fauna are in earlier Ordovician cooler-water, peri-Gondwana faunas, it is thought that the general lowering of temperatures in the Ashgill allowed colonization in hitherto warmer regions (Fortey & Cocks 1988; Owen, Harper & Jia-yu 1991).

Very few palaeomagnetic data are available for this time interval from the massifs comprising Armorica, due mainly to problems of widespread late Palaeozoic overprinting of the magnetic signal which is ubiquitous throughout Hercynian Europe. Only three sets of data with a Q factor ≥ 3 (quality index factor; Van der Voo 1990) have been published. All of these are from the Armorican Massif, but yield apparently contradictory results. Those of Perroud, Bonhommet & Van der Voo (1983) from dolerite sills of the Crozon Peninsula ($Q = 3$) and Ruffet *et al.* (1992) from the Tregor dolerites ($Q = 4$) both indicate intermediate palaeolatitudes. However, in neither of these studies is there good structural or age control, and so neither can be reliably used for any geotectonic interpretation. The results of Perroud & Van der Voo (1985) from the Thouars Massif ($Q = 5$), Vendée, on the other hand, are obtained from well-dated rocks and indicate rather high palaeolatitudes. There is again little structural control for these intrusives, but the original authors state it is unlikely that such a large igneous complex has been tilted to any great degree, and they consider the *in situ* results as representative of the upper Ordovician palaeofield direction in the Armorican Massif.

GEOLOGICAL SETTING AND SAMPLING

The Barrandian Basin forms a synclinal structure within the gently folded Cadomian basement of central Bohemia. It belongs to the Tepla–Barrandian Terrane (Matte *et al.* 1990), forming a relatively stable block between the Moldanubian and Saxothuringian terranes of the Bohemian Massif. Deformation is much less intense in the Tepla–Barrandian than surrounding areas, and it is bounded by deep-seated fault zones (Matte *et al.* 1990) along which major transpressional movement and shearing occurred in Devonian times (Franke 1990). Although the basement to the Barrandian Basin has undergone metamorphism to lower greenschist facies, the lower Cambrian to Givetian basin-fill remains essentially unmetamorphosed. Deformation of the Barrandian Basin sequence is thought to have occurred shortly after the close of sedimentation in the Givetian (Franke 1990). This is in agreement with the late Givetian age of deformation for the surrounding Tepla–Barrandian region as indicated from K–Ar studies (Müller-Sohnius *et al.* 1987).

After peaking in lower Ordovician times, volcanic activity in the Barrandian Basin was fairly restricted. However, pyroclastics and basalt agglomerates of the Komárov Volcanic Complex persisted locally until upper Caradoc times. Sedimentation during the Ashgill resulted in deposition of the Kralov Dvur Formation, a monotonous sequence of grey mudstones with a varied and diversified deep-water fauna. These are overlain by lower Hirnantian diamictite horizons containing faceted pebbles and dropstones (Brenchley & Storch 1989). The diamictites are overlain by a few metres of shales, followed by a sandy shaley sequence belonging to the rhythmically bedded Kosov Formation with hummocky cross-stratification and storm-generated sandstones indicating shallow marine conditions (Storch 1990). Rapid transgression then resulted in the deposition of black graptolite shales which persisted throughout most of the Silurian.

In this study, samples have been collected from the youngest members of the Komárov Volcanic Complex, the Kralov Dvur Formation, the overlying Hirnantian diamictite horizons and from the Kosov Formation (Table 1 and Fig. 1). Also sampled was the 350 m long road exposure of shales and sandstones comprising the lower Caradoc Letná Formation (Table 1). At the Hlasna Treban locality (Fig. 1), the Kosov Formation sandstones have been intruded at their base by diabase sills of Silurian (?) age. A baked contact test was conducted on these sedimentary (three sites) and igneous (four sites) rocks.

The attitude of the volcanic horizons was determined from immediately overlying sediments. The structural attitude and formation to which each site belongs are listed in Table 1. All samples were collected using a petrol-driven rock drill, with orientation by means of a standard sun compass and/or Brunton magnetic compass.

LABORATORY AND DATA SELECTION PROCEDURES

The field-drilled cores were cut into standard 2.2×2.5 cm specimens and magnetizations measured using a two-axis 'Remanomètre à squid RS 01 (LETI, Grenoble) cryogenic magnetometer in Munich, and 2G cryogenic magnetometers in the laboratories at Leoben, Austria, and Zurich, Switzerland. Samples were demagnetized using stepwise thermal techniques, with measurement of the magnetic susceptibility using a KLY2 Kappameter after each heating step.

The demagnetization results were analysed using orthogo-

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Table 1. Formations sampled, stratigraphic age, bedding (azimuth and dip), number of samples measured for each site, and the category to which each belongs (see text for explanation).

Category 1			Category 2		
site	bedding	N	site	bedding	N
Siluro/Devonian (?) Sill					
BER4	332/20	8			
BER5	332/20	8			
BER6	332/20	5			
BER7	332/20	6			
Kosov Formation - U Ashgill (Hirnantian)					
BER1	332/22	10	BNK2	261/40	9
BER2	332/20	6	LTN3	336/35	5
BER3	332/20	5	LEV1	094/33	9
BNK1	261/40	10	LEV2	094/33	6
LTN1	336/35	7			
LTN2	336/35	7			
LTN4	336/35	11			
Kralov Dvur Formation - U Ashgill (Hirnantian)					
			DVU1	120/28	5
			DVU2	120/28	12
			DVU3	094/22	12
Komarov Complex - U Caradoc					
PRS1	120/38	7	ZDI1	139/21	8
PRS2	120/38	7	ZDI2	139/21	7
PRS3	120/38	8	ZDI3	139/21	6
			ZDI4	139/21	6
Letna Formation - Caradoc					
ZBR1	347/74	12	ZBR2	347/74	6
			ZBR3	347/74	7
			ZBR4	347/74	6

nal vector plots (Zijderveld 1967) and stereographic projections, with linear demagnetization trajectories being defined by at least three successive data points which were selected by eye. The directions of the linear segments were calculated using the principle component analysis of Kirschvink (1980), and statistics calculated in all cases according to Fisher (1953).

PALAEOMAGNETIC RESULTS

Based on the behaviour of the samples during thermal demagnetization, the samples collected in this study can be grouped into essentially two categories (see Table 1) which do not appear to be directly related to lithological type. The first corresponds to samples which are either magnetically unstable yielding no reliable magnetic vector, or which reveal apparently colinear vector plot segments but do not exhibit any directional coherence at site or formation level. The second category comprises directions of magnetization which are coherent at site and formation level, and are significantly different from the present-day field direction *in situ*. Although the 'success' rate for ancient magnetization identification was rather poor (approximately 40 per cent), we are confident both that no palaeomagnetic information can be gleaned from the group 1 samples, and that the magnetizations obtained from group 2 (from the Komarov, Kosov and Letna Formations) are ancient (Palaeozoic) in origin as discussed below.

Komarov Volcanic Complex

Two directions of magnetization can be identified during thermal demagnetization of these samples (20 samples, three sites). After removal of a present-day component below 250 °C, a well-defined component with fairly narrow unblocking temperature spectra up to 560 °C can be identified (Fig. 2). This south-easterly and intermediate negative direction *in situ*, and south-westerly steep negative direction after bedding correction (Table 2), is termed the C magnetization. Isothermal remanent magnetization (IRM) and subsequent demagnetization experiments indicate magnetite as the remanence carrier.

Kosov and Letna Formations

Samples from these formations comprise Caradocian sandstones and shales, and upper Ashgillian sandstones, diamictites and volcanics. The volcanic rocks from site LTN4 yield stable high unblocking temperature C directions of magnetization between 400 and 550 °C, often after removal of a present-day overprint. Samples from site LTN1 show slightly different behaviour. The lower temperature component of magnetization persists until temperatures of 480 °C, but it does not correspond to either a present-day or recent overprint (Table 2), and cannot be identified in samples from any of the other sites. The second, higher unblocking temperature component is revealed between 480 and 600 °C with a steep northerly and negative *in situ* direction of magnetization. Despite the uncertainties with regards to the lower blocking temperature component from this site, the stable endpoint direction is considered reliable and is termed the C magnetization (Table 2).

Of the sites collected from the sedimentary rocks of these formations, four yielded reliable palaeomagnetic vectors. Of these, most show the presence of a present-day or recent overprint at lower unblocking temperatures (<300 °C). Upon further thermal demagnetization of samples from the diamictite beds, and the Kosov sandstones (not affected by sill emplacement), a magnetization direction, which is consistent within site and has fairly broad unblocking temperature spectra (300–600 °C), can be identified (Fig. 3). Although the *in situ* directions for this component, termed the C magnetization, are rather dispersed at the site level, correcting for structural attitude brings them into good agreement with each other and with the directions identified in the volcanic rocks (Table 2). Magnetite and minor amounts of haematite are present in the sediments as potential remanence carriers, and many also show the presence of pyrrhotite. This is considered to be of secondary origin and appears to carry the low blocking temperature (<300 °C) component.

Baked contact test

At Hlasna Treban (locality BER, Fig. 1), the Kosov sandstones have been intruded at their base by a diabase sill. The base of the sill is not exposed, but it has a minimum thickness of 25 m. In order to conduct a baked contact test, the sites within the sandstones are located 0.5 m (BER3), 15 m (BER2) and 30 m (BER1) above the baked margin.

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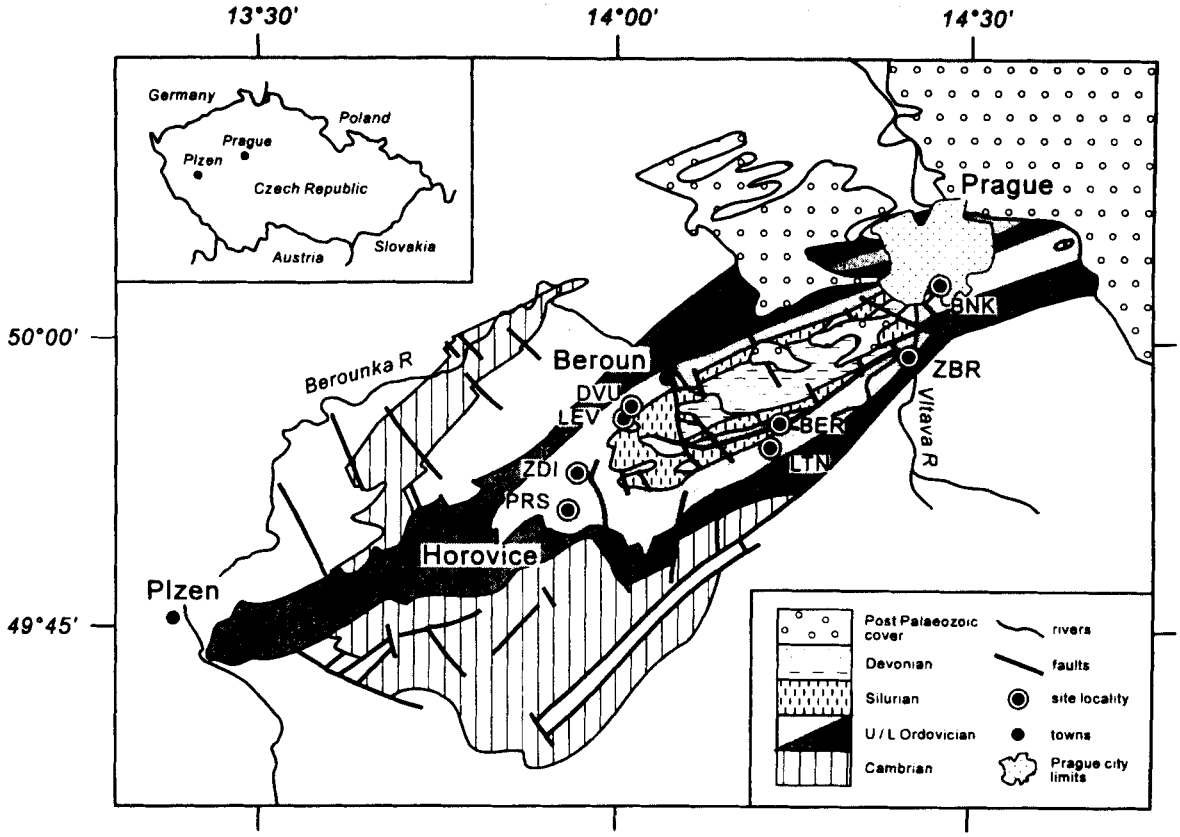


Figure 1. Geological sketch map of the Barrandian Basin, and its location within central Europe. Stars indicate the sampling localities.

Samples were also collected from within the sill itself (Table 1). During thermal demagnetization, the sandstone samples reveal the presence of up to three directions of magnetization, often after removal of a present-day overprint at low temperatures. These are, a southerly and shallow direction of magnetization, a north/north-easterly intermediate positive direction of magnetization, termed direction B, and a steeply negative direction. This latest component is only identified in the highest unblocking temperature range in sites located furthest from the contact (BER1). In the site adjacent to the contact (BER3), the B magnetization completely dominates and has an unblocking temperature spectrum of 20–550 °C. Samples from the sill itself are very stable during thermal demagnetization; a present-day overprint is removed below 250 °C, above which component B is revealed and persists until complete demagnetization at 550–600 °C. Magnetite is the predominant remanence carrier in both the igneous and

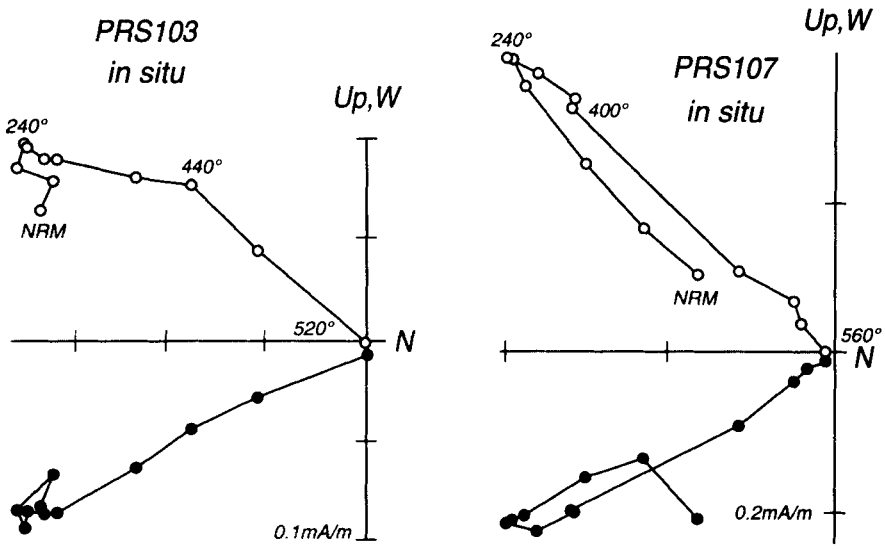


Figure 2. Orthogonal projection of thermal demagnetization results from samples of the Komarov Volcanic Complex. Closed (open) symbols represent the horizontal (vertical) components.

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Table 2(a). Site mean directions for component B.

site	Lithology	N/n	in situ Dec/Inc	k	α_{95}	bedding corrected Dec/Inc	k	α_{95}
BER1	sandstone	10/0	-/-	-	-	-/-	-	-
BER2	sandstone	6/6	011/+31	13.3	19.1	006/+15	13.3	19.1
BER3	baked sandstone	5/5	013/-40	367.1	4.0	005/+23	367.1	4.0
BER4	diabase	8/7	359/+42	109.1	5.8	354/+24	109.1	5.8
BER5	diabase	8/8	350/+39	38.3	7.0	348/+22	38.3	7.0
BER6	diabase	5/5	341/+34	68.8	9.3	339/+15	68.8	9.3
BER7	diabase	6/6	004/-46	85.3	8.3	356/-29	140.0	7.8
BNK1	diamictite	10/0	-/-	-	-	-/-	-	-
LTN1†	diabase	7/7	001/-33	193.5	4.4	025/−61	193.5	4.4
LTN2	diamictite	7/0	/	-	-	-/-	-	-
LTN4	diabase	11/0	-/-	-	-	-/-	-	-
ZBR1	sandstone	12/0	-/-	-	-	-/-	-	-
PRS1	basalt agg	7/0	-/-	-	-	-/-	-	-
PRS2	basalt agg	7/0	-/-	-	-	-/-	-	-
PRS3	basalt agg	8/0	-/-	-	-	-/-	-	-
mean		6 sites	360/+39	52.3	9.3	355/+21	52.3	9.3

Basalt agg, basalt agglomerate; N, number of samples measured; n, number of samples used in calculation of site mean; † site not used in calculation of mean.

sedimentary rocks. In the sediments furthest from the contact small amounts of haematite are also present, but in those adjacent to the sill the magnetic properties are more akin to those of the igneous rocks themselves.

DISCUSSION OF PALAEOMAGNETIC RESULTS

Two ancient components of magnetization can be reliably identified from these late Ordovician rocks. Component B is identified at Hlasna Treban, in samples of the Kosov Formation which are closest to the sill where it represents a local remagnetization event related to sill emplacement. It is also identified in sites from the sill itself as the (presumed) primary direction. The overall mean direction obtained for this magnetization is 360°/39°, $\alpha_{95} = 9.3$, $k = 52.3$ *in situ*. However, in the absence of either field tests or radiometric ages for the sill, the age of this magnetization remains unclear. On geological and petrological grounds, the sill has been interpreted as being of Siluro–Devonian age. If this is the case, then applying a bedding correction (folding occurred in late Devonian to early Carboniferous times) yields an overall mean direction of 355°/21°, $\alpha_{95} = 9.3°$, $k = 52.3$. This is clearly different from the late Silurian palaeomagnetic direction of 208°/−40° previously obtained from the Barrandian Basin by Tait *et al.* (1994b). No reliable palaeomagnetic results have been published for the Devonian of the Bohemian Massif, and the only published data for Hercynian Europe as a whole is that of Bachtadse, Heller & Kröner (1983). The direction of 184°/19° predicted from the palaeopole of Bachtadse *et al.* (1983) is significantly different from the observed B direction (355°/21° after bedding correction). However, the disparity is in terms of declination only. Therefore, a possible explanation is that the B magnetization does correspond to this Devonian palaeopole position, but that large-scale rotations have occurred between this area and the rest of Hercynian Europe since acquisition of the magnetization.

Table 2(b). Site man directions for component C.

site	N/n	in situ Dec/Inc	k	α_{95}	bedding corrected Dec/Inc	k	α_{95}
BER1	10/7	198/-74	41.5	9.5	172/-55	41.5	9.5
BNK1	10/7	034/+49	28.7	11.5	334/+61	28.7	11.5
LTN1	7/5	334/-67	138.6	6.5	158/-78	138.6	6.5
LTN2	7/5	247/-58	34.2	13.3	204/-45	34.2	13.3
LTN4	11/8	245/-72	32.4	9.9	185/-51	32.4	9.9
ZBR1	12/8	139/+63	28.7	11.5	003/-40	28.7	11.5
PRS1	7/6	146/-41	102.1	6.7	188/-69	102.1	6.7
PRS2	7/7	150/-42	109.9	5.8	196/-67	109.9	5.8
PRS3	8/8	151/-37	222.4	3.7	188/-64	222.4	3.7
mean	9 sites	192/-71	6.0	23.0	183/-59	30.3	9.5
palaeopole					80°S;360°E		

Direction C is identified in volcanic and sedimentary rocks as linear segments trending towards the origin in orthogonal projection. The positive contact test for this component (from sites BER1–7) implies that no remagnetization has occurred since sill emplacement, and the statistical improvement in the grouping of site mean directions after bedding correction (192°/−71°, $\alpha_{95} = 23.0$, $k = 6.0$ *in situ*, 183°/−59°, $\alpha_{95} = 9.5°$, $k = 30.3$ after bedding correction; Table 2 and Fig. 4) is significant at the 99 per cent level of confidence applying the fold test of McElhinny (1964). This component, therefore, is interpreted as being closely bracketed with the rock age and being representative of the late Ordovician palaeofield direction in the Bohemian Massif. Using the normal polarity option, the site mean direction yields a palaeopole position of 80°S; 360°E which translates into palaeolatitudes of 40°S for the Bohemian Massif in Caradoc/Ashgill times.

Adopting this polarity option implies large-scale (approximately 170°) rotation of the Bohemian Massif before

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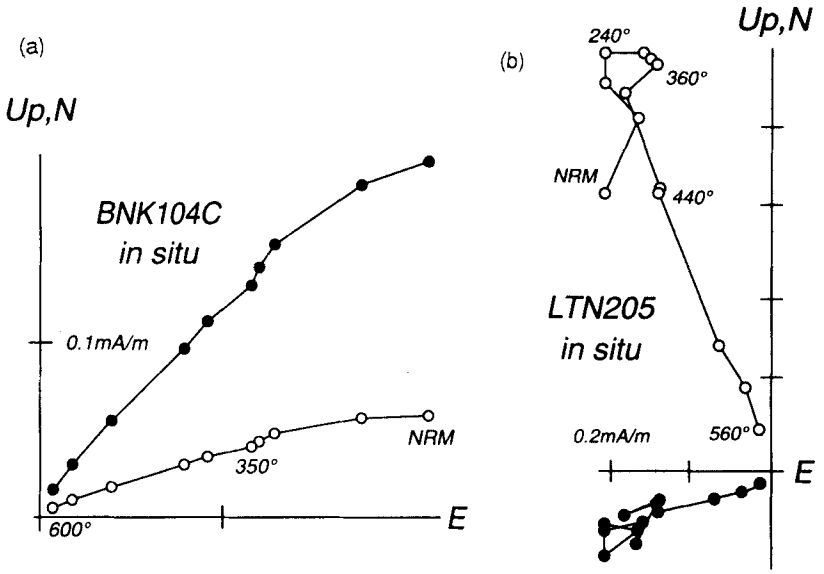


Figure 3. Thermal demagnetization characteristics of (a) sandstone and (b) diamictite beds of the Kosov Formation. Notation is as for Fig. 2.

final consolidation of Hercynian Europe. This implied rotation is consistent with previous results from lower Ordovician and upper Silurian rocks of the Barrandian Basin (Tait *et al.* 1994a, b). In the absence of reliable palaeomagnetic data for this time period from elsewhere in Hercynian Europe, it is not clear whether these rotations are applicable for Armorica as a whole, or if they relate merely to the Bohemian Massif. The latter option is preferred, primarily because such large amounts of rotation are more easily accommodated on the smaller scale. The actual geodynamic mechanism for the rotation is unclear. If, as proposed by Young (1990), the Bohemian Massif was situated at the leading edge of Armorica, the various elements of which formed a longitudinally elongate domain, then oblique collision and differential movement between this element and Baltica may provide a possible explanation. There is some geological evidence for this hypothesis: the Moldanubian/Saxothuringian boundary to the north of the Bohemian Massif is a zone along which major dextral transpressional movement occurred in Devonian times. Furthermore, the Devonian was a period during which a number of small oceanic basins opened and closed within the Hercynian framework (Matte 1991), which would allow for differential movement of the various elements.

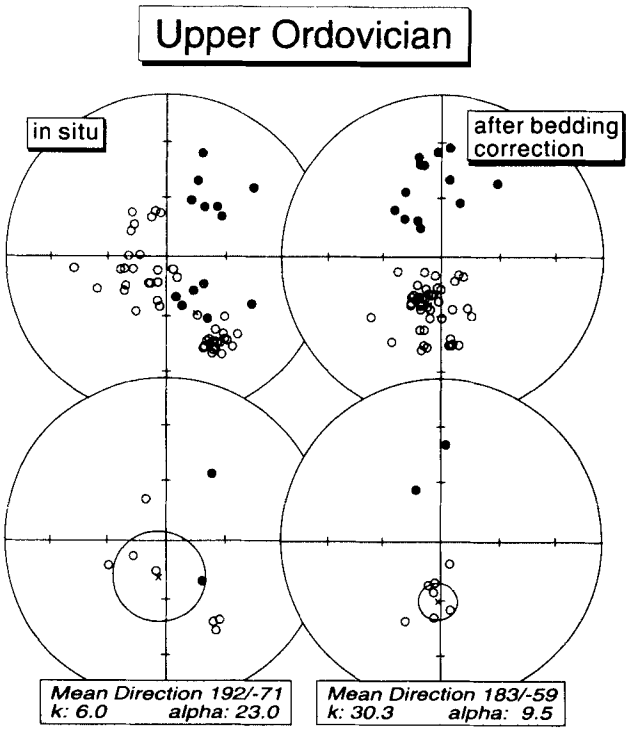


Figure 4. Equal-area stereographic projection of *in situ* and bedding-corrected samples and site mean directions, where solid (open) symbols represent lower (upper) hemisphere projections.

CONCLUSIONS

From the results presented in this paper, the palaeogeographic position of the Bohemian Massif in latest Ordovician times is now fairly well constrained. Although the palaeopole position cannot be used in the construction of an APW path due to the rotations involved, it does demonstrate that by the Ashgillian, the Bohemian Massif had drifted northwards and was located at 40°S (Fig. 5). This corroborates the view that the presence of the *Hirnantia* faunas and Hirnantian glaciomarine deposits in the Bohemian Massif reflects a temporary but widespread lowering of annual average temperatures, affecting areas far removed from the polar ice-caps, which allowed these faunas and lithologies to develop in hitherto warmer regions (Fortey & Cocks 1988; Owen *et al.* 1991).

As already mentioned, the only other reliable palaeomagnetic data available for the upper Ordovician of Armorica are those of Perroud & Van der Voo (1985) from the Thouars Massif. The original authors assume that this igneous complex has not been tilted since emplacement, and the results indicate high palaeolatitudes for the Armorican Massif (approximately 75°S). Two scenarios are possible to

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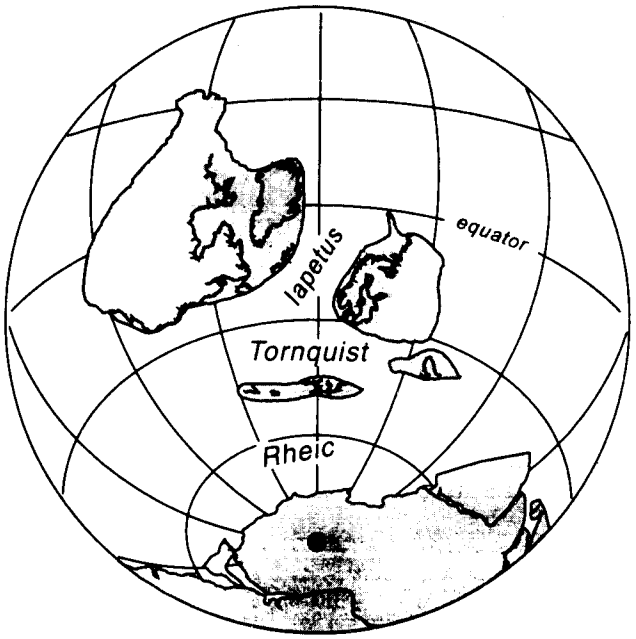


Figure 5. Palaeogeographic reconstruction for upper Ordovician times using data from this study (Armorica), Channell & McCabe (1992) (Avalonia), Torsvik, *et al.* (1992) (Baltica), Van der Voo (1993) (Laurentia) and Bachtadse & Briden (1990) (Gondwana).

explain the apparent discrepancy between these data: either the results of Perroud & Van der Voo (1985) are valid as given and indicate separation of some 3000 km between the Armorican and Bohemian Massifs in the upper Ordovician, or alternatively the Thouars Massif has indeed been tilted, in which case the Armorican and Bohemian Massifs could have been situated at similar palaeolatitudes. Faunal and lithological indicators are in favour of the latter hypothesis, as there is no evidence for any major separation of these terranes, but until more palaeomagnetic data are forthcoming it is not possible to say which hypothesis is correct.

Within a wider context, these new results have important implications for the geodynamic evolution of the Iapetus bordering continents as a whole. They demonstrate that, by the upper Ordovician, both Avalonia and Armorica had rifted from the northern margin of Gondwana and they were situated in more temperate palaeolatitudes. However as stated earlier, palaeomagnetic, lithological and faunal data indicate an earlier separation of Avalonia from Gondwana (Arenig/Llanvirn) than Armorica which remained at high palaeolatitudes until at least the lower Llanvirn, and that Avalonia and Armorica remained distinct until late Silurian times.

The position of Avalonia in the upper Ordovician is rather unclear from palaeomagnetic data. Full discussion of the data set for Avalonia is outwith the scope and aims of this paper, but it is clear that between the Llandeilo and eastern Wenlock it drifted from 45°S (Trench & Torsvik 1991) to 13°S (Torsvik *et al.* 1993). Details of its drift in the intervening period, however, remain ambiguous. Two early studies of the Borrowdale Volcanics, northern England, which are stratigraphically constrained as latest Llanvirn to early Ashgill in age, suggest palaeolatitudes of 15°S (Faller 1977; Piper 1979), placing Avalonia adjacent to Baltica. However, the analytical techniques and selection criteria employed in these studies do not stand up to modern reliability requirements. In a recent study of this volcanic complex, Channell & McCabe (1992) isolate a secondary magnetization direction which indicates palaeolatitudes of 43°S which is in agreement with Siluro–Ordovician results from western Avalonia (Johnson & Van der Voo 1990). The remagnetization in the Borrowdale Volcanics is thought to be related to late stage or syn-volcanic hydrothermal activity and as such is closely bracketed with the rock age, i.e. latest Llanvirn to early Ashgill. Given the ambiguity within the palaeomagnetic data set, and the eastern Wenlock palaeolatitude of 13°S, it has been argued that shallower palaeolatitudes are a more likely scenario (see Torsvik *et al.* 1993), but in the absence of any more reliable palaeomagnetic data, the reconstruction shown in Fig. 5 is based on the results of Channell & McCabe (1992) and Johnson & Van der Voo (1990). Whichever scenario is valid, it is now clear that Avalonia and Armorica were situated at temperate/subtropical palaeolatitudes in the Ashgill. Around this time, there was also a slight increase in the faunal similarities between Armorica and Avalonia, and both domains show an influx of Baltic-type faunas. It is not suggested that Avalonia and Armorica formed a coherent tectonic element as there are clear differences between them, both faunally and from palaeomagnetic data as mentioned earlier. Armorica continued to move northwards during the Silurian, and by the latest Silurian it had collided with the southern margin of Baltica (Tait *et al.* 1994b), shortly after the lower Silurian closure of the Tornquist Sea between Avalonia and Baltica (Torsvik *et al.* 1993).

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